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National Institute of Open Schooling (NIOS)
Senior Secondary Course
Lesson – 13: Cartesian System of Rectangular Co-ordinates
Worksheet -13

1. If three vertices of a rectangle ABCD as $A(0,-1), B(6,7), C(-2,3)$, then find the co-ordinate the fourth vertex D.

2. Draw Cartesians co-ordinate axis on the plan paper. Identify four quadrants and take any two points on any quadrant.

3. If the co-ordinates of the mid-points of the sides of a triangle are (1, 1), (2, –3) and (3, 4), then find its centroid.

4. Find the co-ordinates of the point which divides the line segment joining the points (3, –4) and (–5, 7) internally and externally in the ratio 5:3.

5. Take vertices of a rectangle and show that figure formed by joining the midpoints of the sides of a rectangle is a rhombus.

6. Taking randomly the co-ordinates of vertices of any triangle ABC, find the area of the triangle ABC.

7. For what value of *x*, three points $(2,3),(x,5)$ and (5, 12) are collinear?

8. If a line is equally inclined the axes, prove that slope is positive one or negative one.

9. Find the point's one x-axis, where distance from the line is $\frac{x}{3}+\frac{y}{4}=1$ are 4 units.

10. Find the equation of the line perpendicular to the line $x-7y+5=0$ and having x-intercept 3.